

COVER SHEET

SUMMARY

SYMBOLS

LEGEND

POLARITY DISTRIBUTION

INCOMING CHARGE

MANUAL CIRCUIT BREAKER

INCOMING BATTERY

MANUAL CIRCUIT BREAKER

BUS BAR COUPLING

MANUAL SWITCH

OUTGOING

MANUAL CIRCUIT BREAKER

SWITCH POSITION
AND FUSE BLOW

GROUPED FAULT

09

AS BUILT

N.	Date				
N.	Date				
N.	Date				
N.	Date				
D	N.	DUR	HAR	DEG	AS BUILT
	Date	95/08/09	95/08/09	95/08/09	ASB
C	N.	XAL	HAR	DEG	UP DATED ACCORDING TO MANUFACTURING & GOOD FOR EXECUTION
	Date	94/01/26	94/01/26	94/01/26	GFE
B	N.	XFN	HAR	DEG	UP DATED ACCORDING TO EX02 AND MANUFACTURER DATAS
	Date	93/02/26	93/02/26	93/02/26	AFTER RECEIPT
A	N.	XTO	HAR	DEG	
	Date	92/12/3	92/12/3	92/12/3	ORIGINAL ISSUE
REV.		ETABLI ESTABLISHED	VERIFIE CHECKED	APPROUVE APPROVED	MODIFICATIONS
					STAT.

VINH-SON HYDRO-POWER STATION

2 X 40 MVA

CONTRACT N° 12-05/CEG-PC3-PET/01-91

SOCIALIST REPUBLIC OF VIET-NAM
MINISTRY OF ENERGY
PETECHIM-POWER COMPANY Nr 3
ENGINEER EDF



LEADER **CEGELEC** FRANCE
ALSTHOM - JEUMONT
CEGELEC NEYRPIIC
SDEM

ECHELLE
SCALE

POWER PLANT
125V LOW VOLTAGE DC SYSTEM
TYPICAL ELECTRICAL DIAGRAM

	DATE	NOM NAME	VISA
ETABLI ESTABLISHED	92/12/3	XTO	
VERIFIE CHECKED	92/12/3	HAR	
APPROUVE APPROVED	92/12/3	DEG	

CEGELEC

Division ERE P : 35, rue d'Alsace F - 92531 Levallois Perret Cedex

CGE VIN20 SLAA-ST001

P2135A30084-D

FOLIO
SHEET
01
FIN
19

	DESIGNATION	F0	REVISION															
			A	B	C	D												
A	COVER SHEET	1	X	X	X	X												
	SUMMARY	2	X	X	X	X												
	SYMBOLS	3	X															
	LEGEND	4	X															
B																		
	POLARITY DISTRIBUTION	7	X	X														
	INCOMING CHARGER	8	X	X														
	MANUAL CIRCUIT BREAKER																	
	INCOMING BATTERY	9	X	X														
	MANUAL CIRCUIT BREAKER																	
C	BUS BAR COUPLING	10	X															
	MANUAL SWITCH																	
	OUTGOING	11	X	X	X													
	MANUAL CIRCUIT BREAKER																	
	SWITCH POSITION	14	X	X														
	AND FUSE BLOWN																	
D	GROUPED FAULT	15	X	X														
	METERING AND PROTECTIVE SYSTEM	18	X	X	X													
E	HEATERS AND LIGHTING CUBICLES	19	X	X														
F																		

SUMMARY

P2135A30084-D

Division
ERE P
02 03

BULZLE1309

ANSI	DESIGNATION	CEI	ANSI	DESIGNATION	CEI
12	OVERSPEED DEVICE	c >	50-76	OVERCURRENT RELAY	I >
13	SYNCHRONOUS-SPEED DEVICE	c c c >	50N	OVERCURRENT RELAY, NEUTRAL	I N >
14	UNDERSPEED DEVICE	>	51	TIME-LAG OVERCURRENT RELAY	I L >
15	SPEED MATCHING DEVICE	c c c >	51	INVERSE TIME OVERCURRENT RELAY	I L >
21	DISTANCE OR UNDERIMPEDANCE RELAY	Z >	51N	TIME-LAG OVERCURRENT RELAY, NEUTRAL	I N >
21	DISTANCE OR UNDERREACTANCE RELAY	x >	51V	TIME-LAG OVERCURRENT RELAY WITH VOLTAGE RESTRAINT	I > U >
21	DISTANCE OR OVERADMITTANCE RELAY	y >			
23	TEMPERATURE CONTROL DEVICE	θ	53	EXCITATION INCREASE RELAY	B >
25	VOLTAGE COMPARISON DEVICE	u ~	55	POWER FACTOR RELAY	cos φ
25	PHASE-ANGLE COMPARISON DEVICE	u ~	56	EXCITATION RELAY	B
25	FREQUENCY COMPARISON DEVICE	u ~	58	RECTIFICATION FAILURE RELAY	Δ=0
25	COUPLING DEVICE	↑	59	OVERVOLTAGE RELAY	U >
25	SYNCHRONIZING OR SYNCHRONISM-CHECK DEVICE	u f	59N	NEUTRAL OVERVOLTAGE RELAY	U N >
26	THERMOMETER DETECTOR	θ	59-81	OVERINDUCTION RELAY	U / f
27	UNDERVOLTAGE RELAY	u <	60	VOLTAGE BALANCE RELAY	U d
27	NO-VOLTAGE RELAY	u = 0	60	CURRENT BALANCE	I d
30	ANNUNCIATOR RELAY	⊗	63	PRESSURE SWITCH	P
32	DIRECTIONAL ACTIVE POWER RELAY	⊗	64	FRAME LEAKAGE RELAY	f
32	DIRECTIONAL REACTIVE POWER RELAY	⊗	67	DIRECTIONAL OVERCURRENT RELAY	I N
32	REVERSE ACTIVE POWER RELAY	⊗	67N	DIRECTIONAL HOMOPOlar OVERCURRENT RELAY	I N
32	REVERSE REACTIVE POWER RELAY	⊗	71	LEVEL SWITCH	o
37	UNDERCURRENT RELAY	u <	74	ALARM RELAY	⊠
37	UNDER POWER RELAY	u <	78	MEASURING RELAY AND PHASE-ANGLE PROTECTION	e
39	MECHANICAL CONDITION MONITOR	B >	79	RECLOSING RELAY	f
40	EXCITATION FAULT RELAY	B = 0	81	FREQUENCY RELAY	f
40	NO-EXCITATION RELAY	B = 0	87	DIFFERENTIAL PROTECTIVE RELAY	Δ I
46	PHASE-BALANCE OR REVERSE-PHASE CURRENT RELAY	I >	87N	DIFFERENTIAL PROTECTIVE RELAY	Δ I
47	VOLTAGE RELAY	u >	95	BUCHHOLZ RELAY	Δ I
49	THERMAL RELAY	θ			
49	THERMOSTAT	θ			

YU	VOLTAGE MEASUREMENT RELAY
YT	TEMPERATURE MEASUREMENT RELAY
YG	ISOLATION MEASUREMENT RELAY
MU	VOLTAGE TRANSDUCER
MI	CURRENT TRANSDUCER
MW	POWER TRANSDUCER
II	CURRENT INDICATOR
IU	VOLTAGE INDICATOR
BM	TEST BOX
TU	VOLTAGE TRANSFORMER
TI	CURRENT TRANSFORMER
CC	CONTROL SWITCH

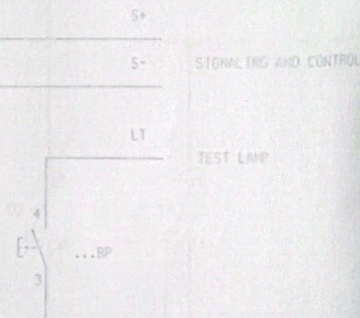
SWITCH	RESIDUAL CURRENT TRANSFORMER	CAPACITOR	MANUALLY OPERATED
DISCONNECTOR	SHUNT	CAPACITOR VOLTAGE DIVIDER	ELECTROMAGNETIC ACTUATOR
CIRCUIT BREAKER	INDICATING INSTRUMENT	INDUCTOR WINDING	PNEUMATIC OR HYDRAULIC CONTROL
CONTACTOR	RECORDING INSTRUMENT	BATTERY	REVERSER
CONTACTOR POLE (contact closed in the unoperated position)	ENERGY METER	CONVERTER OR MEASURING TRANSDUC.	KEY ALWAYS FREE
SWITCH DISCONNECTOR POLE (on load isolating switch)	IMPULSE COUNTER	RECTIFIER	KEY FREE FOR THE POSITION IN THE DIAGRAM
FUSE	MEASURING RELAY OR RELATED DEVICE	PLUG AND SOCKET	KEY IMPRISONED FOR THE POSITION IN THE DIAGRAM
THERMAL EFFECT MAGNETIC EFFECT	SIGNAL LAMP	ELECTRICAL CONTROL	ISOLATED PHASE BUS DUCT
THREE WAYS SWITCH	LIMITER	LINK	CABLE
ROTATING MACHINE	DOUBLE DISCONNECTOR	INVERTER	OVERHEAD LINE
TRANSFORMER WITH 2 WINDINGS	SPARKGAP	EARTH MASS	CARRIER LINE TRAP
TRANSFORMER WITH 3 WINDINGS	LIGHTNING ARRESTER	PULSE	PLUG (male)
AUTO-TRANSFORMER	RESISTOR	MECHANICAL INTERLOCK	SOCKET (female)
CURRENT TRANSFORMER	VARIABLE RESISTOR	KEY OPERATED	POWER LINE CARRIER COUPLING DEVICE

SYMBOLS

Division
ERE P
CEGELEC
P2135A30084-A
FOLD 03
VOIR 04

LEGEND AND ELECTRICAL COMPONENTS

BP	PUSH BUTTON
	ON:GREEN STOP:RED RESET FAULT:YELLOW
	TEST LAMP:BLEU OTHER:WHITE
LA	LAMPS
	ON:RED STOP:GREEN FAULT:YELLOW OTHER:WHITE
FU	FUSES
XR	AUXILIAIRY RELAY
CD	FAULT AUXILIARY CONTACT
CA	POSITION AUXILIARY CONTACT
FCD	DISCONNECTED POSITION CONTACT
FCE	CONNECTED POSITION CONTACT
FC	LIMIT SWITCH
JD	CIRCUIT BREAKER
JC	CONTACTOR
JS	DISCONNECTING SWITCH
EA	TRIPPING COIL
YI	CURRENT MEASUREMENT RELAY
YU	VOLTAGE MEASUREMENT RELAY
YT	TEMPERATURE MEASUREMENT RELAY
YG	ISOLATION MEASUREMENT RELAY
MU	VOLTAGE TRANSDUCER
MI	CURRENT TRANSDUCER
MW	POWER TRANSDUCER
II	CURRENT INDICATOR
IU	VOLTAGE INDICATOR
BM	TEST BOX
TU	VOLTAGE TRANSFORMER
TI	CURRENT TRANSFORMER
CC	CONTROL SWITCH



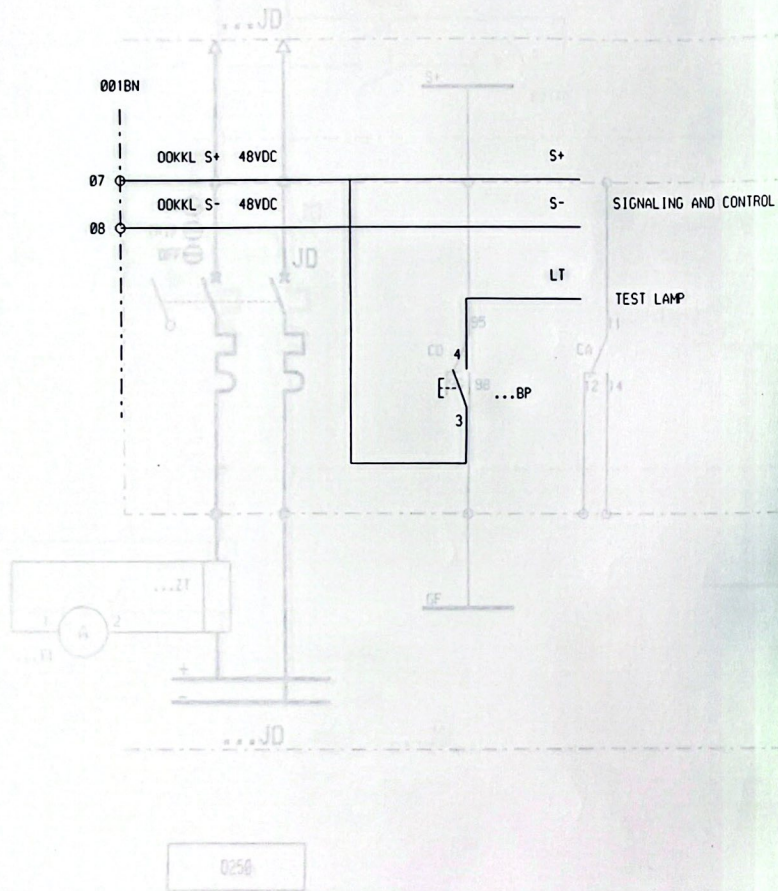
POLARITELEGENDRIBUTION

CEGELEC

Division
ERE P

P2135A30084- A

04 07



INCOMING CHARGER
POLARITY DISTRIBUTION
2 POLES

CEGELEC
P2135A30084 - B

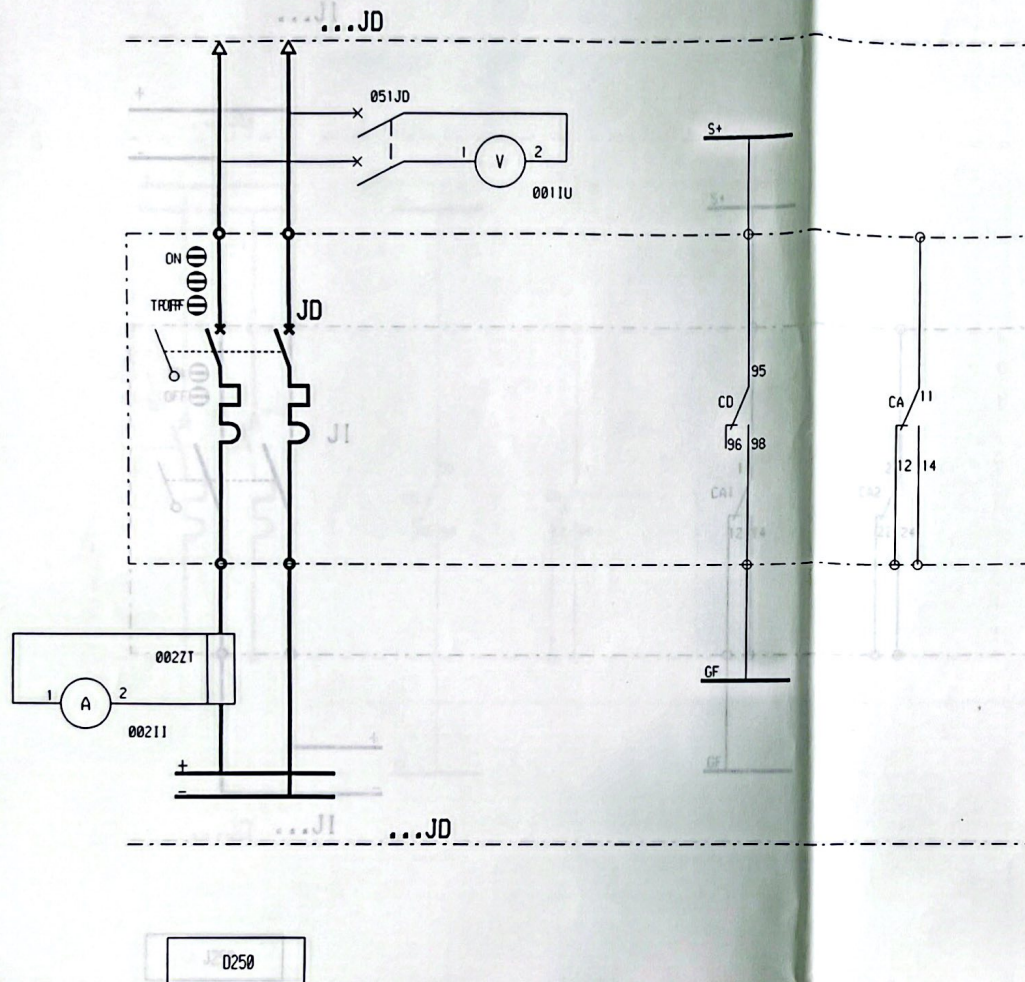
Division

ERE P

FOLD
SHEET
07

WOP
SHEET
08

8 KENCZLE 134



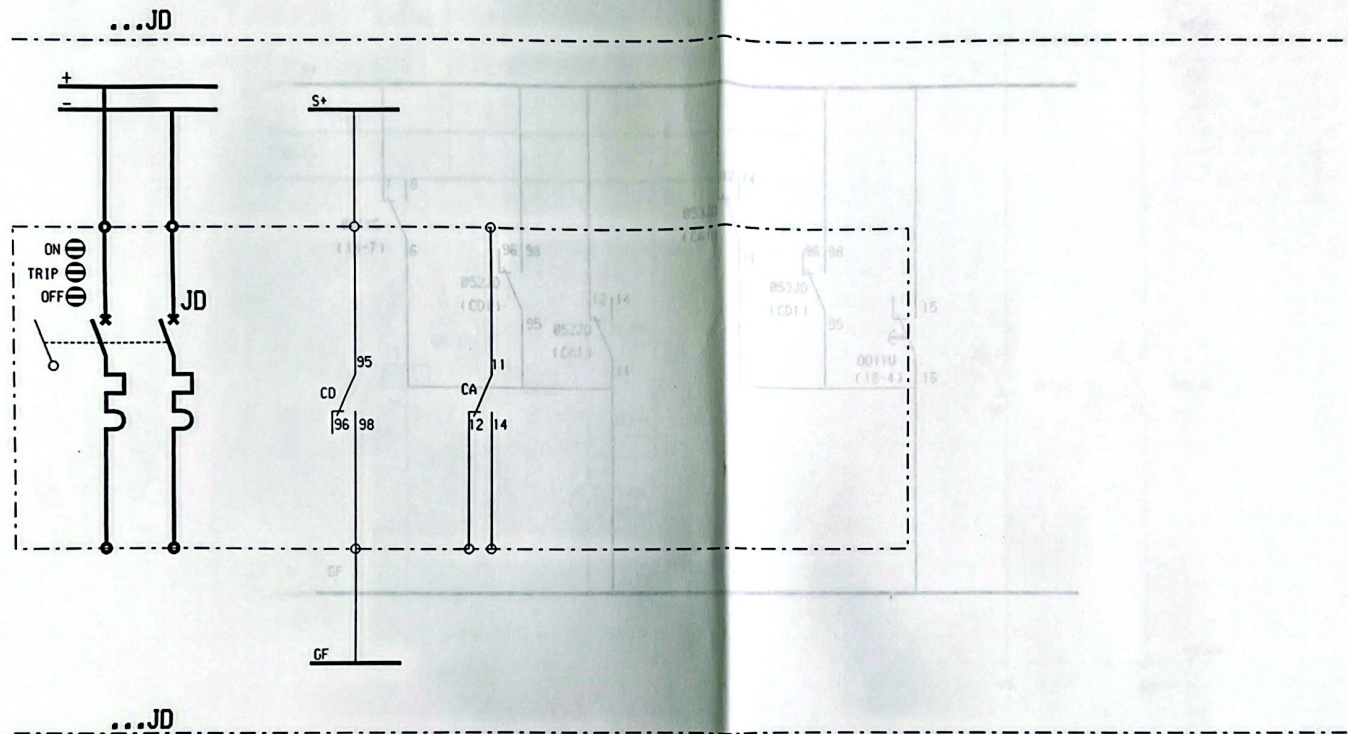
INCOMING BATTERY
MANUAL CIRCUIT BREAKER
2 POLES

CEGELEC
P2135A30084-B

Division
ERE P

EQUIP
09
VOR
10

ENC 21 E 1 34



D100 D160

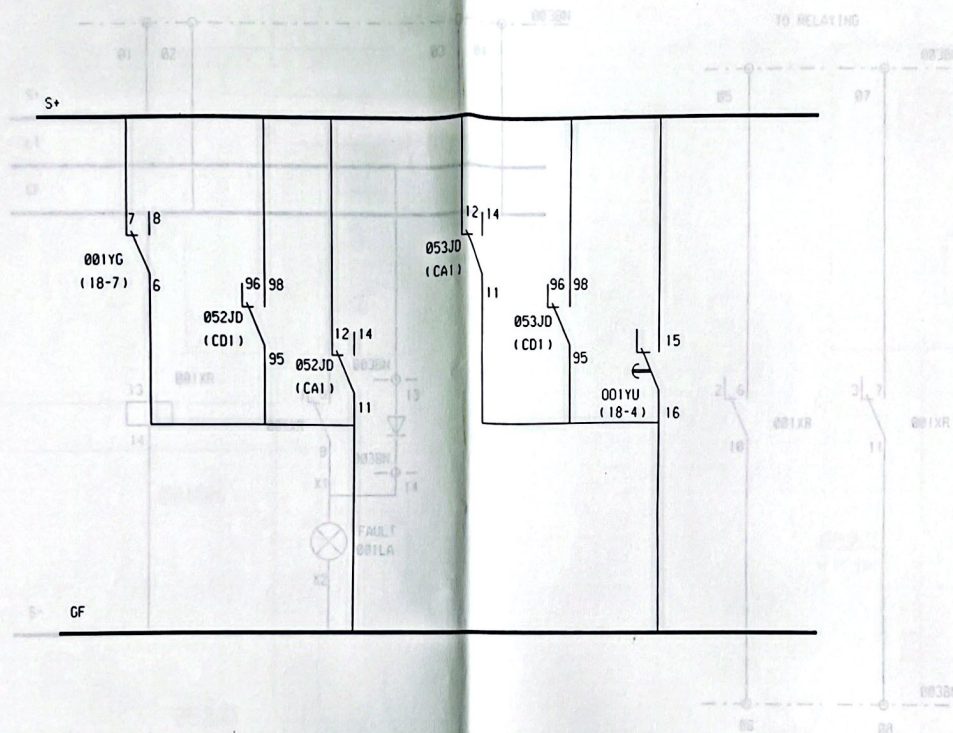
SW OUTGOING TION
MANUAL CIRCUIT BREAKER
INSULATED 2 POLES ROLLER

CEGELEC
P2135A30084-c
Division
ERE P
EOLD
SHEE
11
VOR
14

PERMANENT INSULATED CONTROLLER

UNDERVOLTAGE RELAY

REMOTE FAULT BATTERY CHARGER 1 AND 2



SWITCH POSITION
FUSE BLOWN ☒ UNDERVOLTAGE
INSULATED CONTROLLER

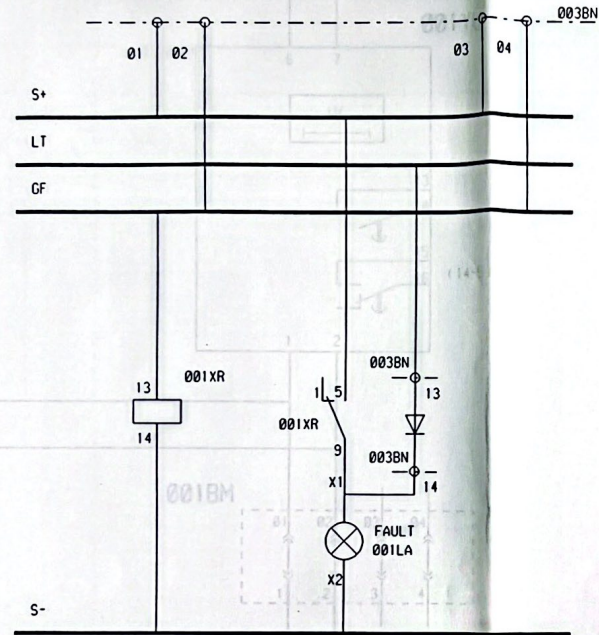
P2135A30084-B

Division
ERE P

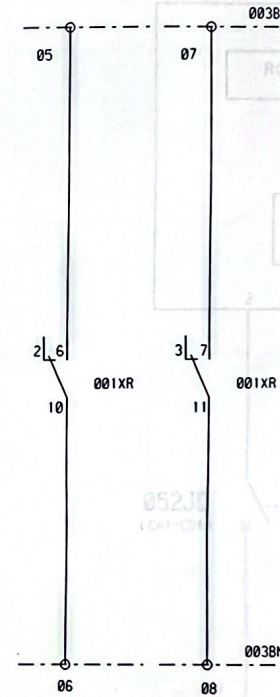
14 15

X10021E1332

REMOTE FAULT BATTERY CHARGER 1 AND 2



TO RELAYING



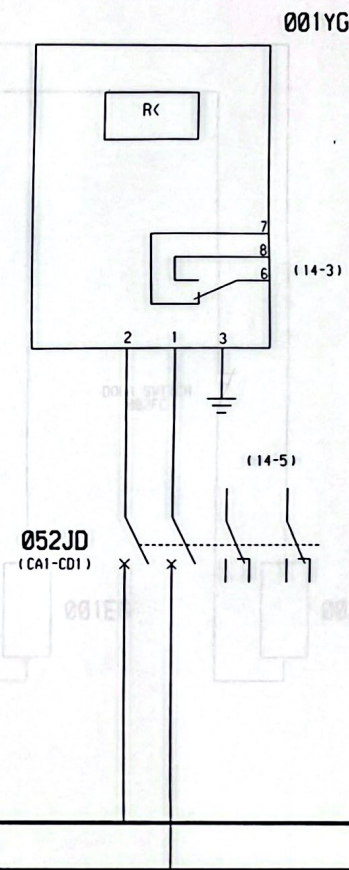
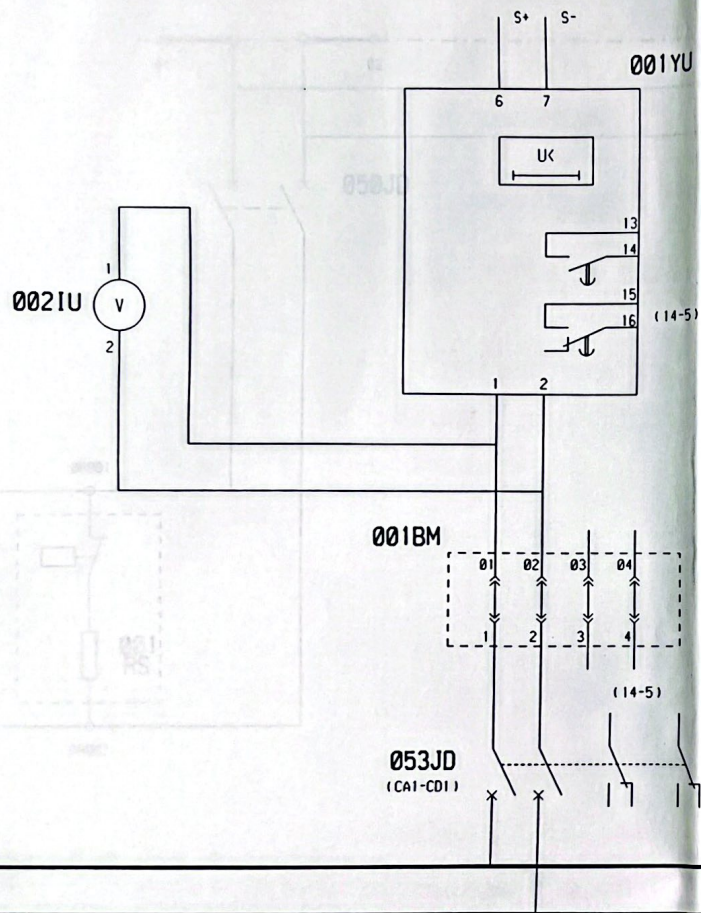
GROUPED FAULT
SYSTEM

CEGELEC

Division
ERE P

P2135A30084 - B

FOLD
SHEET 15
VOIR
SHEET 18

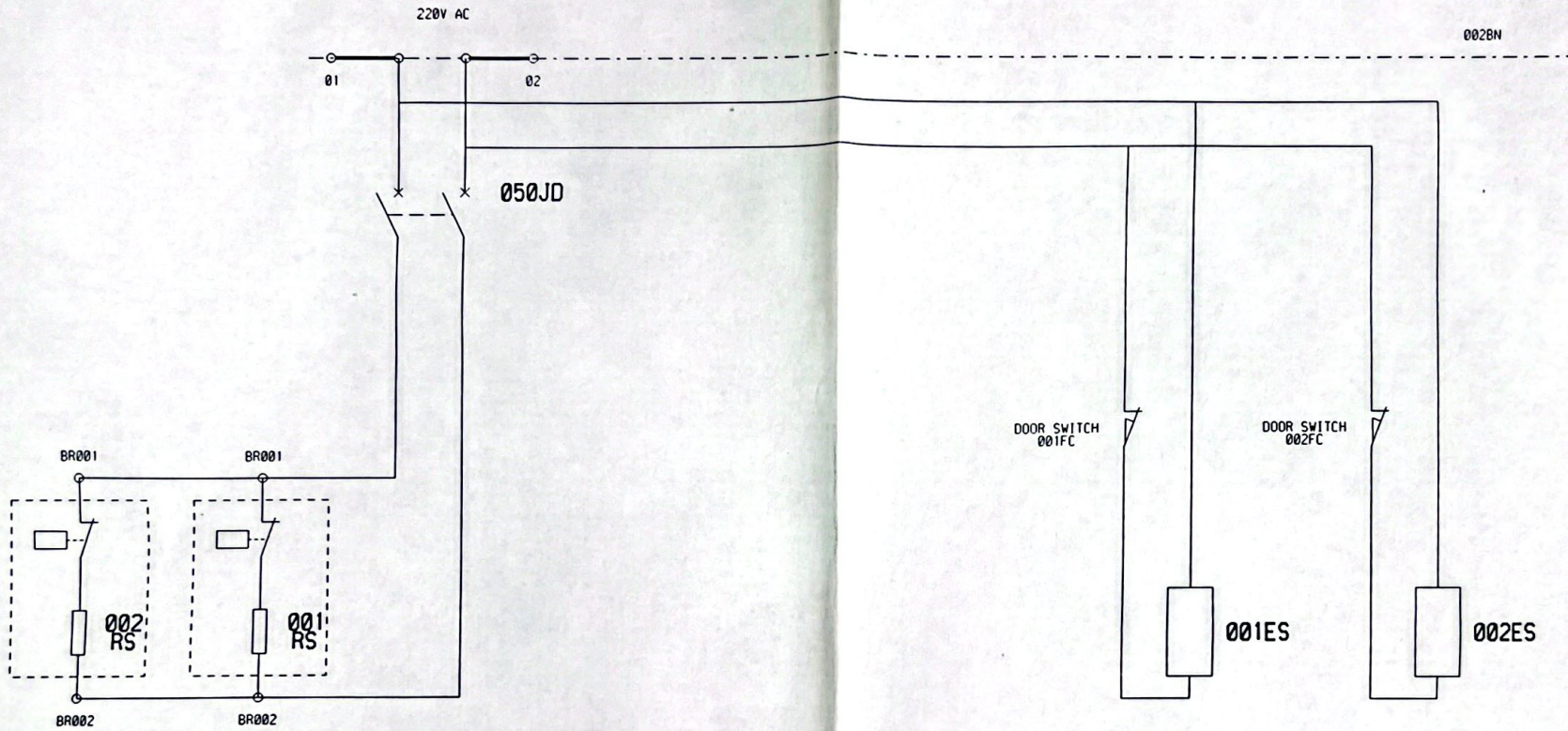


PROTECTIVE AND METERING
SYSTEM

CEGELEC
P2135A30084-c3
Division
ERE P
FOLD
SHEET 18
VOR
SHEET 19

CUBICLES HEATERS *

LIGHTING CUBICLES



HEATERS AND LIGHTING CUBICLES

P2135A30084-B

Division
ERE P

19

134